

Chapter 3: Model Selection in ARMA(p,q) processes

Wold's Decomposition Theorem and the ARMA(p,q) models

- One reason for the popularity of the ARMA models derives from Wold's Decomposition Theorem
- Any covariance stationary process has infinite order, moving-average representation (a linear combination of uncorrelated random variables with mean zero and a deterministic component):

$$X_t = \sum_{i=0}^{\infty} \psi_i L^i e_t + d_t$$

$$e_t \sim iidN(0, \sigma^2)$$

- Note that the Wold decomposition is unique. In other words, if two stationary processes have the same Wold's decomposition, they are the same process.

Wold's Decomposition Theorem and the ARMA(p,q) models

- Let

$$\psi(L) = 1 + \psi_1 L + \psi_2 L^2 + \dots$$

- $\psi(L)$ can be approximate by an ARMA(p,q) process. It is a parsimonious model and a good approximation for a wide class of stationary processes.

$$\psi(L) = \frac{B(L)}{A(L)} = \frac{1 + b_1 L + b_2 L^2 + \dots + b_q L^q}{1 - a_1 L - a_2 L^2 - \dots - a_p L^p}$$

ARMA(p,q) Models

- **Common factor problem:** identification of an ARMA(p,q) model requires the lag polynomial A(L) and B(L) to have **no common factors**, otherwise the system is not identified.

Example: Suppose you have the following ARMA(2,1) model:

$$(1 - 0.25L^2)X_t = (1 + 0.5L)e_t$$

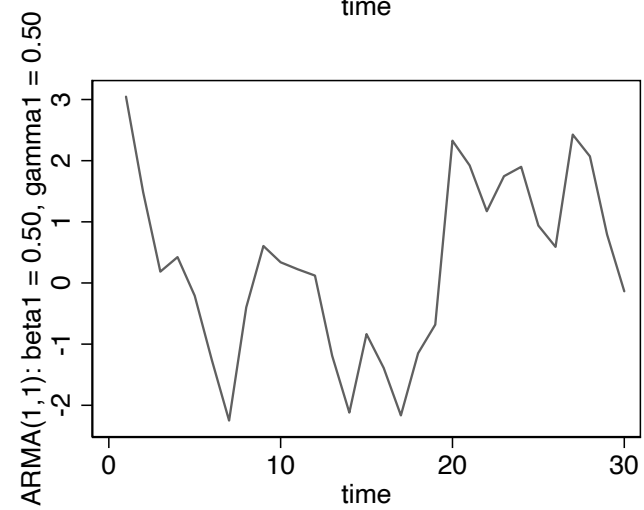
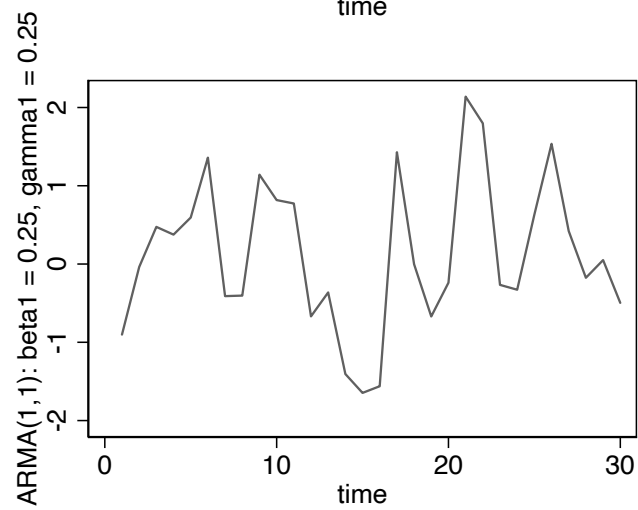
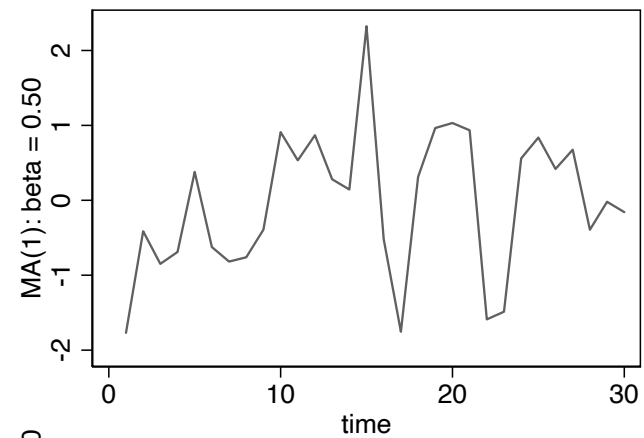
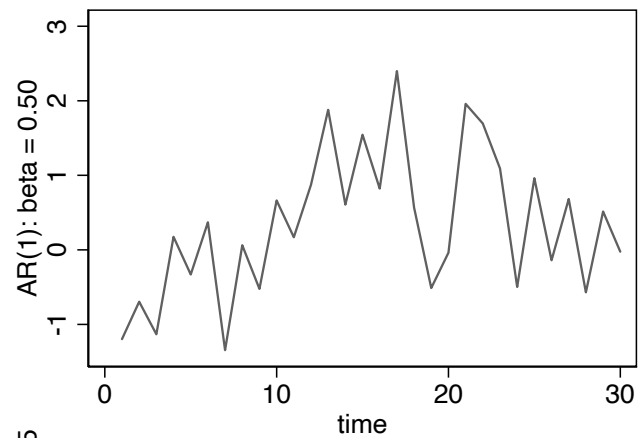
This model has a common factor and can be factored as an AR(1), a more parsimonious form. The common factor $(1 + 0.5L)$ is not identified.

$$(1 - 0.25L^2)X_t = (1 + 0.5L)e_t$$

$$(1 - 0.5L)(1 + 0.5L)X_t = (1 + 0.5L)e_t$$

$$X_t = 0.5X_{t-1} + e_t$$

ARMA(p,q) processes



Model Selection: The Box-Jenkins approach to modelling stationary time series

Steps :

OBS: It is important to make sure that the time series is stationary. If not, transformation should be made (we will discuss non-stationary time series in chapter 5).

- **(1) Identification:** determine p and q (model should be parsimonious)
- **(2) Estimation**
- **(3) Diagnostics tests** to check for model adequacy (check if the residuals are white noise, if not, go back to step 1!!!).
- **(4) Forecast.**

Model Selection: The Box-Jenkins approach to modelling stationary time series

- **Identification:**

- **autocorrelation function (ACF)**
- **partial autocorrelation function (PACF)**
- **Information Criteria (AIC, BIC)**

➤ The estimated ACF and PACF from the data analyzed is compared with the theoretical ACF and PACF.

Theoretical ACFs: AR(1)

X_t is an AR(1) process:

$$X_t = \beta X_{t-1} + e_t$$

$$e_t \sim iid N(0, \sigma^2)$$

$$\text{Cov}(e_t, e_{t-k}) = 0$$

- $E(X_t) = E(X_{t-1}) = 0$ (assuming a zero mean process: $\beta_0 = 0$) and $\text{Var}(X_t) = \text{Var}(X_{t-1})$ (stationary process)
- The autocorrelation of lag 1 (ρ_1) is:

$$\text{Corr}(X_t, X_{t-1}) = \frac{\text{Cov}(X_t, X_{t-1})}{\text{Var}(X_t)} = \rho_1$$

Theoretical ACFs : AR(1)

$$\text{Corr}(X_t, X_{t-1}) = \frac{E(X_t X_{t-1}) - E(X_t)E(X_{t-1})}{E(X_t^2) - E(X_t)E(X_t)} = \frac{E(X_t X_{t-1})}{E(X_t^2)}$$

Computing the $E(X_t X_{t-1})$: multiply both sides of the equation by X_{t-1} and take the expectation:

$$X_{t-1}X_t = \beta X_{t-1}X_{t-1} + e_t X_{t-1}$$

$$E(X_{t-1}X_t) = \text{Var}(X_t) = E(X_t^2) \quad (\text{stationary process})$$

$$\text{Corr}(X_t, X_{t-1}) = \frac{\beta E(X_t^2)}{E(X_t^2)} = \beta$$

Theoretical ACFs : AR(1)

The autocorrelation at further lags will be:

$$\rho_2 = \beta^2$$

$$\rho_3 = \beta^3$$

$$\rho_4 = \beta^4$$

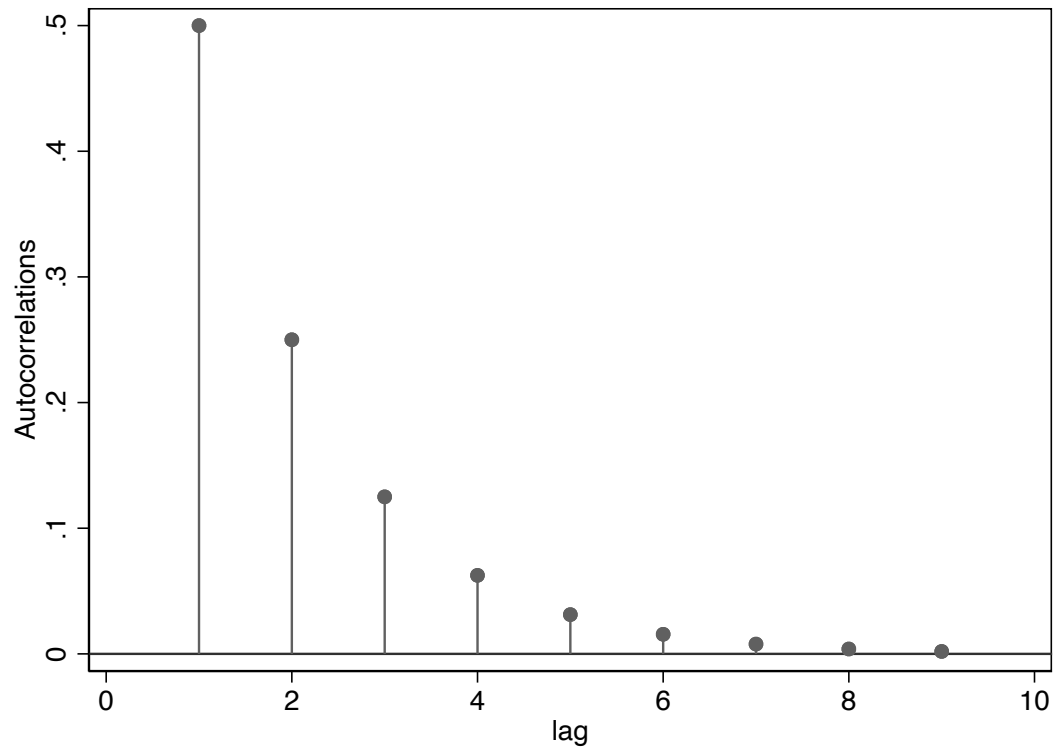
$$\rho_k = \beta^k$$

The theoretical ACF of an AR(1) process **decays exponentially**. If $-1 < \beta < 0$, the ACF will **oscillate between negative and positive numbers, but also will converge to zero**. ρ_k shows the linear dependence between X_t and X_{t-k} .

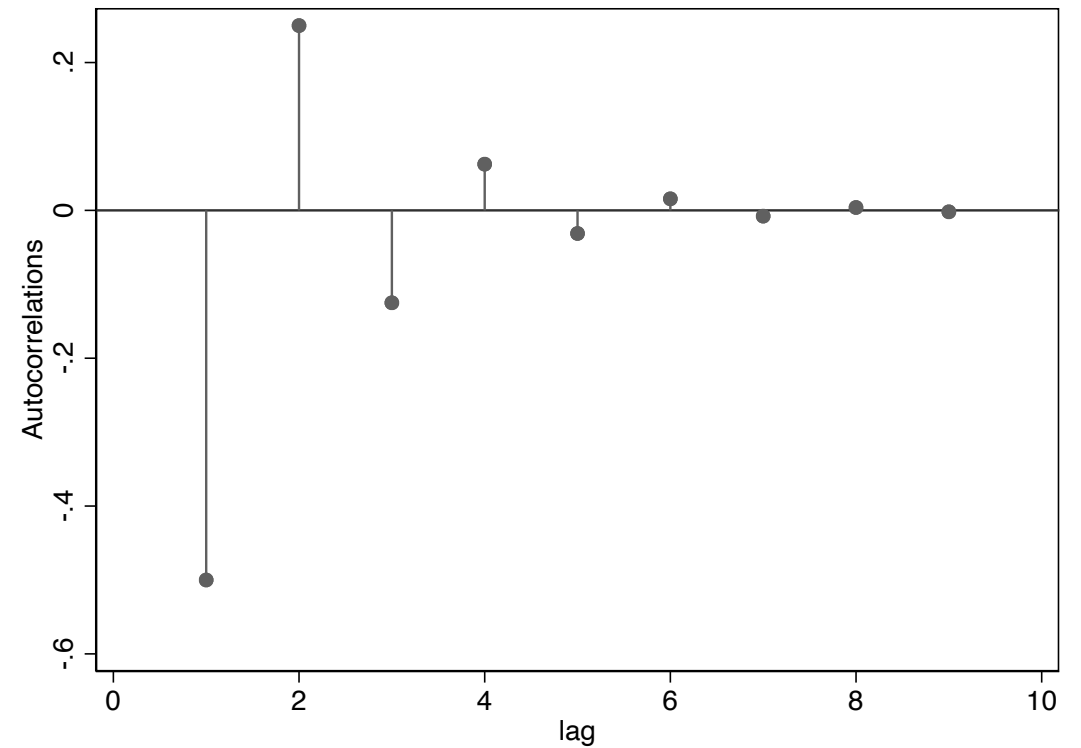
Theoretical ACF of AR(1):

(a) $X_t = 0.5X_{t-1} + e_t$ (b) $X_t = -0.5X_{t-1} + e_t$

(a)



(b)



```
set more off

drop _all

clear all

set obs 1000

set seed 1234

gen time = _n

tsset time

gen X = rnormal()

save "temp_data.dta", replace

* ACF of AR(1)

use "temp_data.dta", clear

constraint define 1 _b[ARMA:L1.ar] = 0.50

quietly arima X , ar(1) noconstant nolog constraints(1)

estat acplot, lags(10) recast(spike) saving("temp", replace)

use "temp", clear

replace lag = lag-1

drop if lag == 0

twoway dropline ac lag, msymbol(O) yline(0)

graph export "theoretical_acf_ar1_a.pdf", replace

erase "temp.dta"

use "temp_data.dta", clear
```

Theoretical ACFs: AR(p)

- plot of the theoretical ACF of a weakly stationary AR(p) process will show a mixture of exponential decays and dampened sine waves.
- Yule-Walker equations:

$$\begin{aligned} \text{Corr}(X_t, X_t) &= \rho_0 = 1 \\ \text{Corr}(X_t, X_{t-1}) &= \rho_1 \\ &\vdots \\ \text{Corr}(X_t, X_{t-k}) &= \rho_k \end{aligned}$$

⇒ are the autocorrelations of an AR(p).

Then the Yule-Walker equations will be:

$$\begin{aligned} \rho_0 - \beta_1 \rho_1 - \beta_2 \rho_2 - \cdots - \beta_p \rho_p &= \frac{\sigma^2}{R(0)} \\ \rho_1 - \beta_1 \rho_0 - \beta_2 \rho_1 - \cdots - \beta_p \rho_{p-1} &= 0 \\ \rho_2 - \beta_1 \rho_1 - \beta_2 \rho_2 - \cdots - \beta_p \rho_{p-2} &= 0 \\ &\vdots \\ \rho_k - \beta_1 \rho_{k-1} - \beta_2 \rho_{k-2} - \cdots - \beta_p \rho_{p-k} &= 0 \end{aligned}$$

Theoretical ACFs: AR(p)

- Example of an AR(2)

$$\rho_0 - \beta_1\rho_1 - \beta_2\rho_2 = \frac{\sigma^2}{R(0)}$$

$$\rho_1 - \beta_1\rho_0 - \beta_2\rho_1 = 0$$

$$\rho_2 - \beta_1\rho_1 - \beta_2\rho_0 = 0$$
$$\vdots$$

$$\rho_k - \beta_1\rho_{k-1} - \beta_2\rho_{k-2} = 0 \text{ for } k \geq 3$$

- Solving for ρ_1 and ρ_2 given that $\rho_0 = 1$, we will have

Theoretical ACFs: AR(p)

$$\rho_1 = \frac{\beta_1}{1 - \beta_2}$$

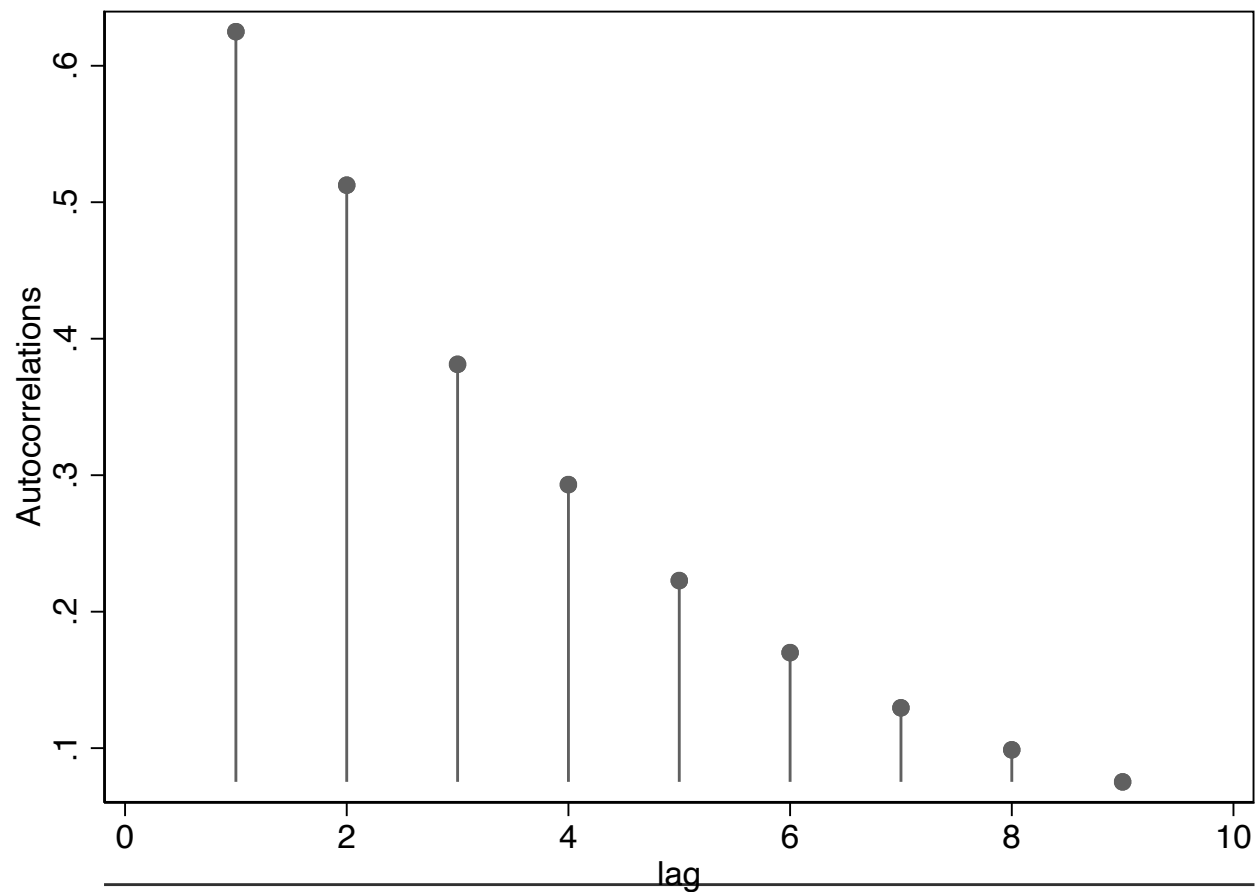
$$\rho_2 = \frac{\beta_1^2}{1 - \beta_2} + \beta_2$$

$$\rho_3 = \beta_1 \rho_2 + \beta_2 \rho_1$$

$\vdots$

$$\rho_k = \beta_1 \rho_{k-1} + \beta_2 \rho_{k-2}$$

Theoretical ACF of AR(2): (a) $X_t = 0.5X_{t-1} + 0.2X_{t-2} + e_t$



Theoretical ACFs: MA(1)

$$X_t = u_t + \beta u_{t-1}$$

$$u_t \sim iid N(\mu, \sigma_u^2) \\ Cov(u_t, u_t) = Var(u_t) = \sigma_u^2$$

$$Cov(u_t, u_{t-j}) = E(u_t \cdot u_{t-j}) = 0, \quad \text{for } t \neq j$$

$$Corr(X_t, X_{t-1}) = \frac{Cov(X_t, X_{t-1})}{Var(X_t)} = \rho_1$$

$$Var(X_t) = Var(u_t) + \beta^2 Var(u_{t-1}) + 2 Cov(u_t, u_{t-1}) = (1 + \beta^2) \sigma_u^2$$

$$\text{Recall that } E(X_t) = 0$$

Theoretical ACFs: MA(1)

$$\text{Cov}(X_t, X_{t-1}) = E(X_t, X_{t-1}) - E(X_t) E(X_{t-1})$$

$$\text{Cov}(X_t, X_{t-1}) = E[(u_t + \beta u_{t-1})(u_{t-1} + \beta u_{t-2})]$$

$$\text{Cov}(X_t, X_{t-1}) = E(u_t \cdot u_{t-1} + \beta u_t \cdot u_{t-2} + \beta u_{t-1}^2 + \beta^2 u_{t-1} \cdot u_{t-2}) = \beta \sigma_u^2$$

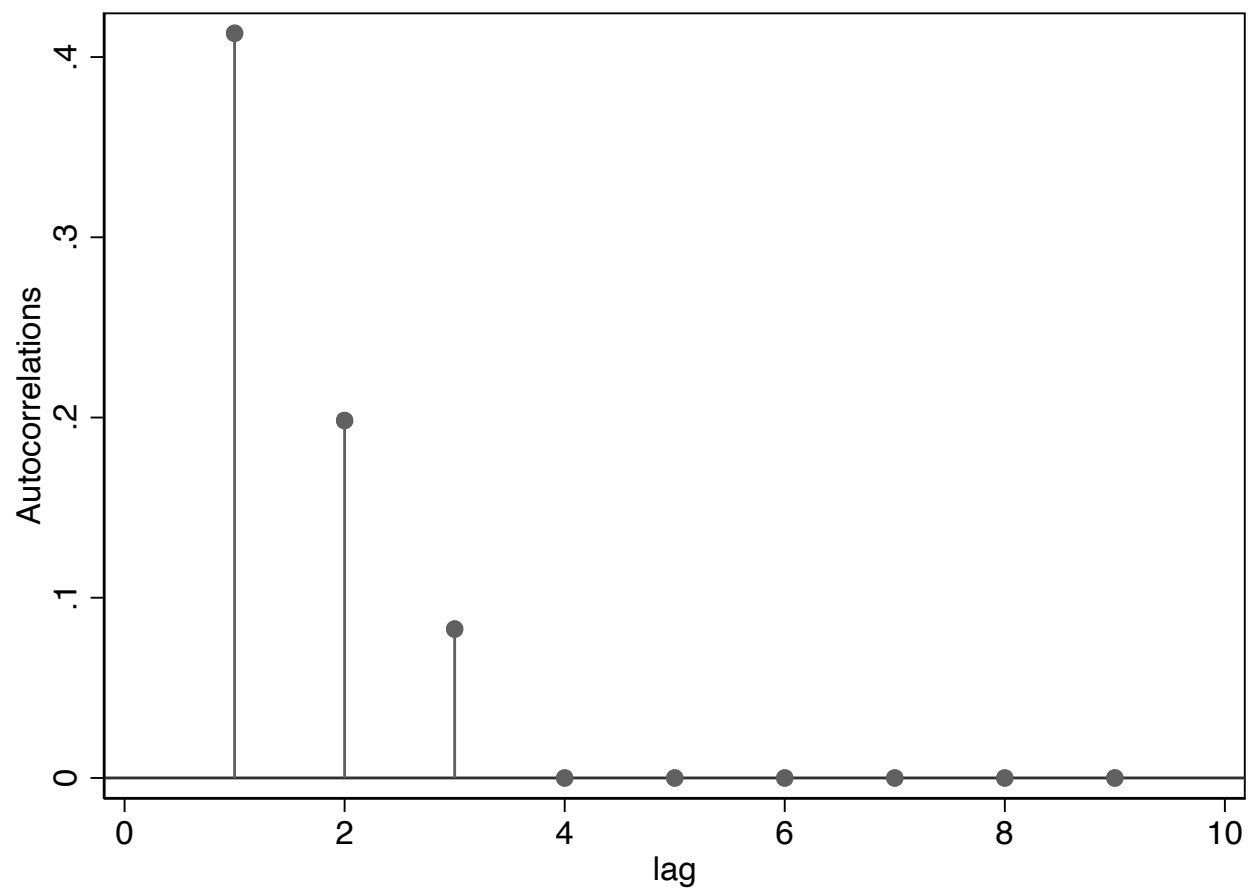
$$\text{Corr}(X_t, X_{t-1}) = \frac{\beta \sigma_u^2}{(1 + \beta^2) \sigma_u^2} = \frac{\beta}{(1 + \beta^2)}$$

And

$$\text{Corr}(X_t, X_{t-k}) = 0, \text{ for } k \geq 1$$

The ACF of and MA (q) dies out after lag q (the order of the MA process)

Theoretical ACF: $X_t = u_t + 0.4u_{t-1} + 0.2u_{t-2} + 0.1u_{t-3}$



Theoretical PACFs

- The partial autocorrelation is the autocorrelation between X_t and X_{t-k} after their mutual linear dependency on the intervening variables $X_{t-1}, X_{t-2} \cdots X_{t-k-1}$ has been removed:

$$\text{Corr}(X_t, X_{t-k} | X_{t-1}, X_{t-2} \cdots X_{t-k+1})$$

(we use the Yule-Walker equation to find the theoretical PACF).

- Theoretical PACFs are the mirrored opposites of ACFs:
 - The ACF of an AR(p) dies down exponentially; the PACF has spikes at lag 1 through p and then it is equal to zero at lags greater than p;
 - The ACF of an MA(p) has non-zero spikes up to lag q and it is equal to zero afterward; the PACF dies down towards zero and has some oscillation.

Theoretical PACFs: AR(1)

- In a AR(1) the autocorrelation of order 1, ρ_1 , is equal the partial autocorrelation of order 1, ϕ_{11} and it is equal zero for $k > 1$.

$$\rho_1 = \phi_{11}$$

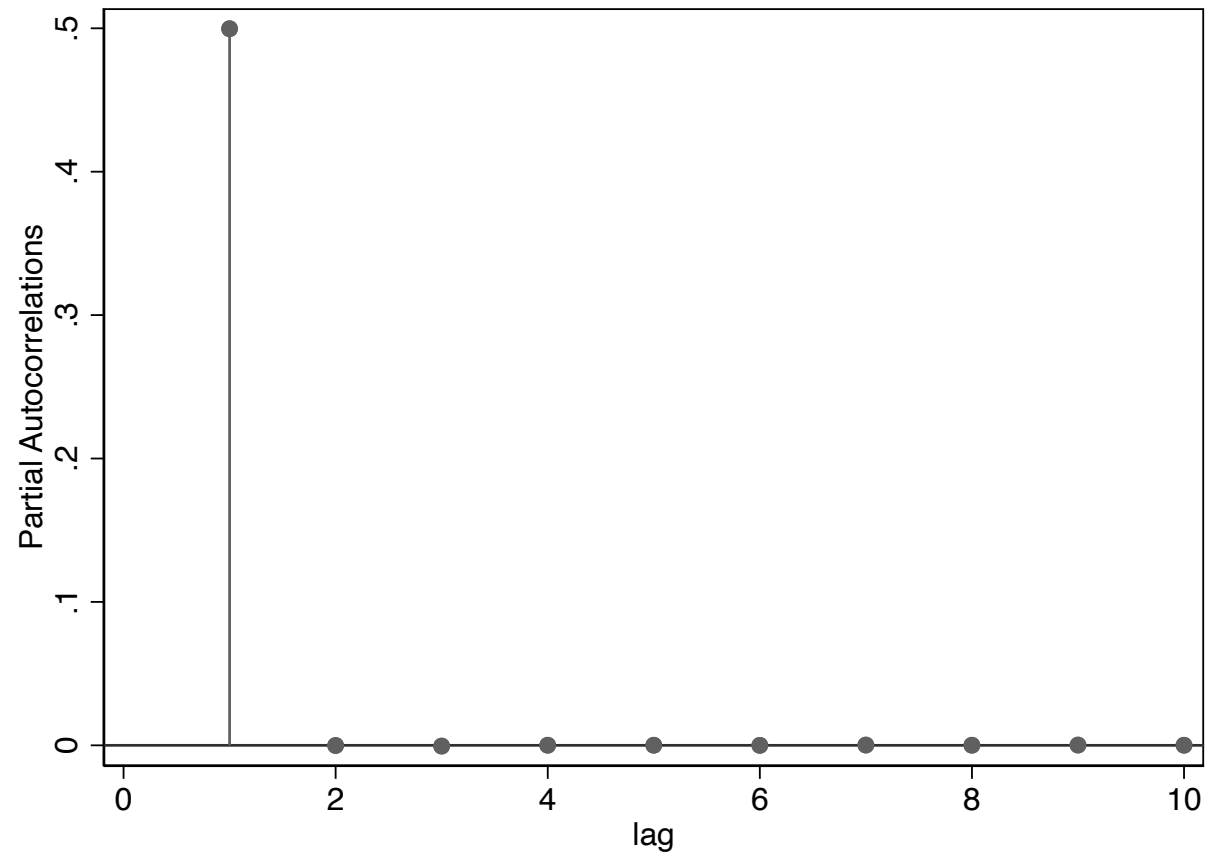
- The partial autocorrelation between X_t and X_{t-2} will be zero. Let $\dot{X}_t = X_t - \mu$

$$\dot{X}_t = \phi_{11}\dot{X}_{t-1} + e_t$$

$$\dot{X}_t = \phi_{21}\dot{X}_{t-1} + \phi_{22}\dot{X}_{t-2} + e_t$$

ϕ_{22} is the partial autocorrelation coefficient between X_t and X_{t-2} , or, the ϕ_{22} is the correlation between X_t and X_{t-2} controlling the effect of X_{t-1} . For an AR(1) there is no direct correlation between X_t and X_{t-2} , then $\phi_{22} = 0$. We can use the Yule-Walker equations to solve for the partial autocorrelation: $\phi_{22} = \frac{\rho_2 - \rho_1^2}{1 - \rho_1^2}$. In an AR(1), $\rho_2 = \rho_1^2$.

Theoretical PACF: $X_t = 0.5X_{t-1} + e_t$



PROPERTIES OF THE ACF AND PACF

(table from Enders)

PROCESS	ACF	PACF
White Noise	All $\rho_k = 0$ ($s \neq 0$)	All $\phi_{kk} = 0$
AR(1): $\beta > 0$	Direct exponential decay: $\rho_k = \beta^k$	$\phi_{11} = \rho_1$; $\phi_{kk} = 0$ for $k \geq 2$
AR(1): $\beta < 0$	Oscillating decay: $\rho_k = \beta^k$	$\phi_{11} = \rho_1$; $\phi_{kk} = 0$ for $k \geq 2$
AR(p)	Decays toward zero. Coefficients may oscillate.	Spikes through lag p. All $\phi_{kk} = 0$ for $k > p$
MA(1): $\gamma > 0$	Positive spike at lag 1. $\rho_k = 0$ for $k \geq 2$	Oscillating decay: $\phi_{11} > 0$
MA(1): $\gamma < 0$	Negative spike at lag 1. $\rho_k = 0$ for $k \geq 2$	Geometric decay: $\phi_{11} < 0$
ARMA(1,1) $\beta > 0$	Exponential decay beginning at lag 1. Sign $\rho_1 = \text{sign}(\beta + \gamma)$	Oscillating decay beginning at lag 1. $\phi_{11} = \rho_1$
ARMA(1,1) $\beta < 0$	Oscillating decay beginning at lag 1. Sign $\rho_1 = \text{sign}(\beta + \gamma)$	Exponential decay beginning at lag 1. $\phi_{11} = \rho_1$ and $\text{sign}(\phi_{kk}) = \text{sign}(\phi_{11})$
ARMA(p,q)	Decay (either direct or oscillatory) beginning at Lag q.	Decay (either direct or oscillatory) after lag p.

EMPIRICAL ACFs and PACFs

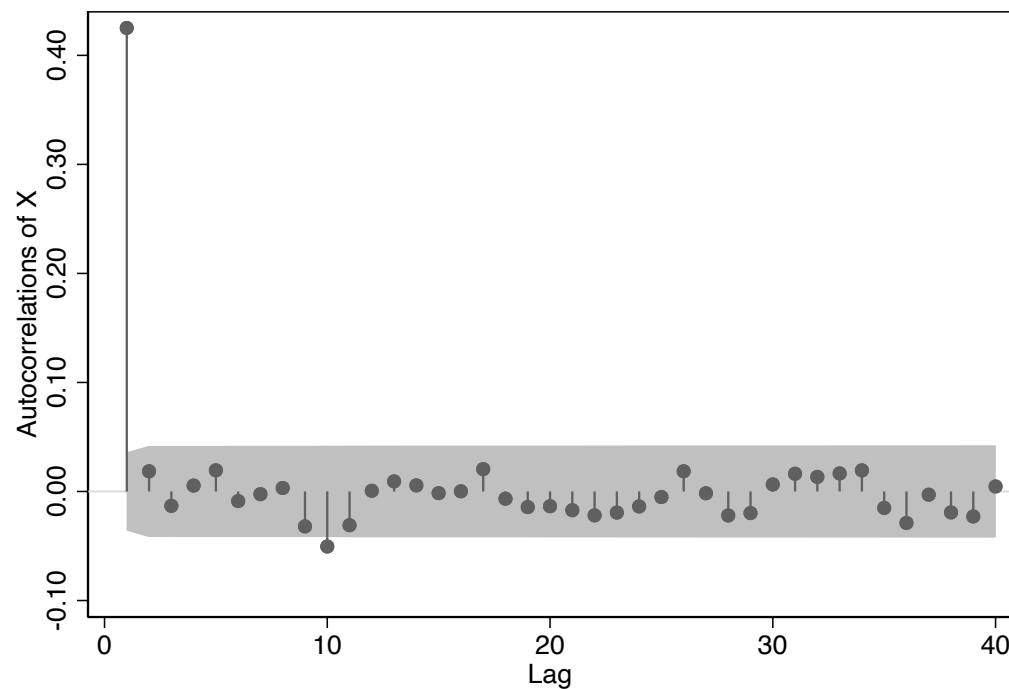
- The empirical ACFs and PACFs are estimated from data. We compare the empirical ACF and PACF with various theoretical ACF and PACF in order to identify the order of the ARM(p,q) model(see guidelines on table above).
- We use the sample values and test if they are significant (a 95% confidence interval is display in the correlogram). We can also use the Ljung and Box statistic to jointly test the null that all values are equal zero against the alternative that at least one autocorrelation is different than zero (the Box-Pierce statistic, known as Q-stat, has a chi-square distribution with s degrees of freedom:

$$Q = T(T + 2) \frac{\sum_{k=1}^s r_k^2}{(T - k)}$$

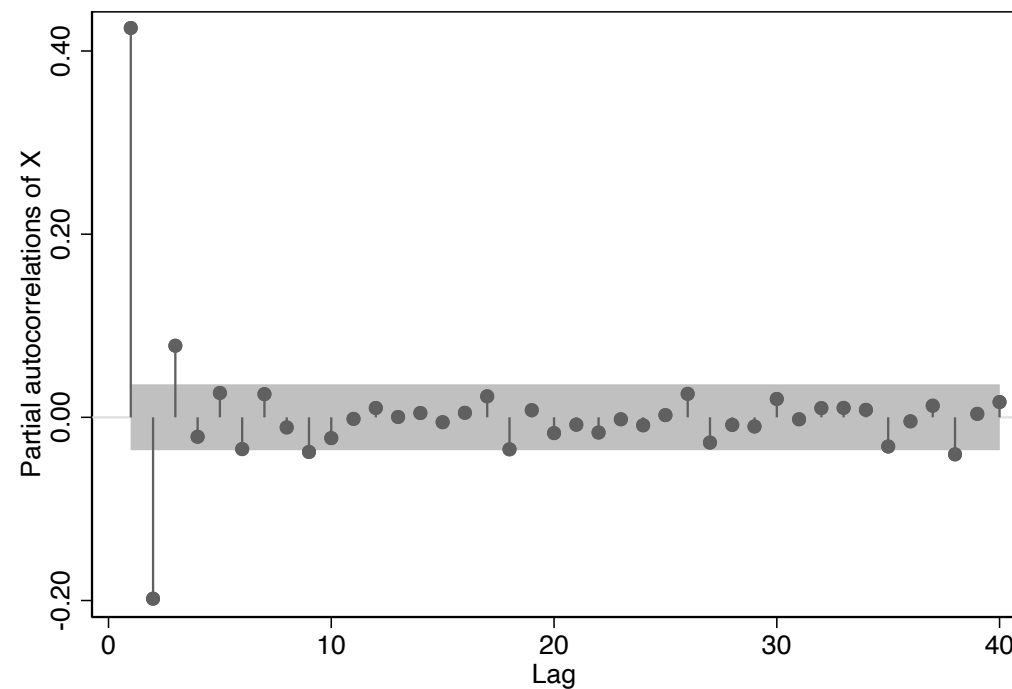
Where r_k are the estimated autocorrelations.

- We can easily compute and graph the estimated function on STATA: **ac** command and **pac** command.
- (show example on page 73 and 74)

Empirical ACF and PACF: $X_t = 0.5u_{t-1} + u_t$

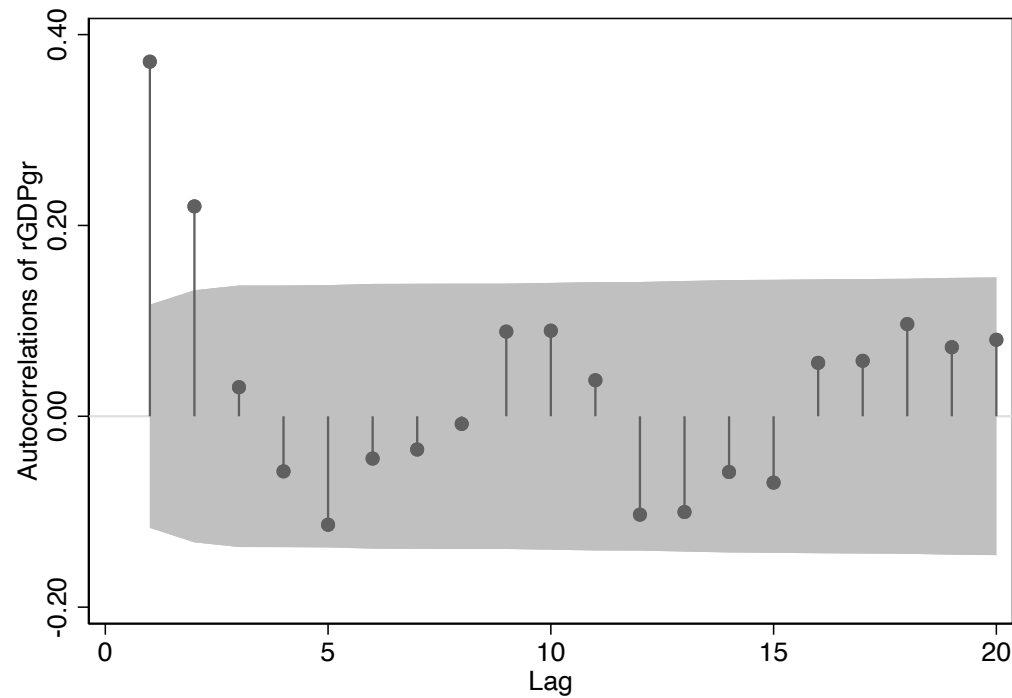


Bartlett's formula for MA(q) 95% confidence bands

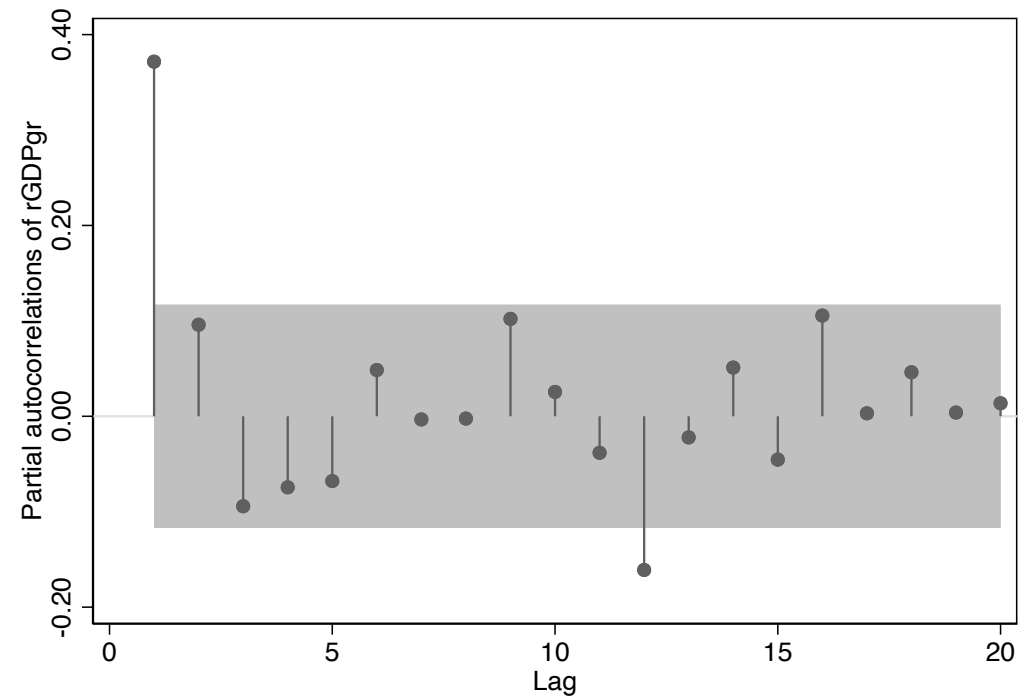


95% Confidence bands [se = 1/sqrt(n)]

Example: Seasonally Adjusted GDP growth rates (quarterly data)



Bartlett's formula for MA(q) 95% confidence bands



95% Confidence bands [se = 1/sqrt(n)]

INFORMATION CRITERIA

- There exists various information criteria. The AIC and BIC are the most commonly used ones.
- Akaike Information Criterion (AIC)

$$AIC = -2\ln(L) + 2k$$

- Schwartz Bayesian Criterion (BIC)

$$BIC = -2\ln(L) + k\ln(N)$$

Where $\ln(L)$ is the maximized log-likelihood of the model and k is the number of parameters estimated (to compare model, we need to use the same N , the number of usable observations).

INFORMATION CRITERIA

- Basically, the information criteria penalize the log-likelihood (the penalty will depend on the number of observations and the number of estimated statistics). For best fitted models, the AIC and BIC will be the smallest as possible (tends to negative infinite).
- The BIC criterion is asymptotically consistent and select a more parsimonious model; the AIC is biased toward selection an overparameterized model. For small sample AIC works better than BIC. The best is if both criteria selected the same model. After the model selection, we should check if the residuals are white noise and the significance of the t-stat of all estimated coefficients of the model .

INFORMATION CRITERIA: Example

- Using the Arexamples. dta

Estimate AR(3), AR(2), AR(1), MA(3), MA(2), and MA(1 for the time series X.

Using AIC and BIC, the **smaller information criteria** will indicate the best fitted model.

Best fitted model: AR(1)

X is in fact and AR(1) process.